TUNDUMA TOWN COUNCIL
SUMAGHA SECONDARY SCHOOL
FORM FOUR MOCK EXAMINATION — MARKING SCHEME
142/1 INFORMATION AND COMPUTER STUDIES
PAPER 1 — THEORY (Marking Guide)

SECTION A (20 Marks)

1. Multiple choice (1 mark each = 10 marks)

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	C	C	A	C	A	C	B	B	C

2. Matching (2 marks each = 10 marks)

– (i) Virus → C

– (ii) Protocol → A

– (iii) Firewall → D

– (iv) ROM → F

– (v) Primary key → E

SECTION B (45 Marks — best 3 of 5 questions, 15 marks each)

3. Computer networks (15 marks)

– (a) Computer network = two or more computers linked together to share data/resources. *(2)*

– (b) LAN — small area e.g. one building/school lab; MAN — covers a city/town e.g. linking council offices in a town; WAN — covers a country/globally e.g. the Internet. 3 marks per network correctly distinguished with example. *(9)*

– (c) Any 4 of: resource sharing (printers/internet), fast communication, centralized data storage/backup, cost saving, easy software updates, collaborative work. 1 mark each. *(4)*

4. Database management systems (15 marks)

– (a) DBMS = software that allows creation, storage, retrieval and management of data in a database. *(2)*

– (b) Any 4 of: reduces data redundancy, ensures data consistency/integrity, allows multiple users to access data, improved data security, faster data retrieval, easy backup and recovery. 2 marks each. *(8)*

– (c) Table = collection of related records; Field = a single category of data/column; Record = a complete set of related fields/row; Primary key = field that uniquely identifies each record. *(5)*

5. Systems Development Life Cycle (15 marks)

– (a) SDLC = a structured step-by-step process/framework followed when developing an information system. *(2)*

– (b) Any 4 stages: Feasibility study, Analysis, Design, Development/Implementation, Testing, Maintenance — each with a one-line explanation. 2 marks each. *(8)*

– (c) Any 3 of: to determine if the project is technically/financially/operationally possible, to avoid wasting resources on an unworkable system, to identify risks early, to estimate cost and time realistically. Accept any reasonable justification. *(5)*

6. Web design / HTML (15 marks)

– (a) HTML = language for creating web pages; Tag = a keyword enclosed in angle brackets that defines an element (e.g.); Attribute = extra information added inside a tag (e.g. href); Hyperlink = a clickable link to another page/resource. *(6)*

– (b) Award marks for correct structure: <html><head><title>...</title></head><body><h1>My School</h1> ...description... </body></html>. 2 marks for <h1> heading, 2 marks for paragraph, 2 marks for correct overall tag structure/nesting. *(6)*

– (c) Any 3 of: a website is a collection of related webpages while a webpage is a single page; a website has a homepage, a webpage doesn't necessarily; a website is identified by a domain name, a webpage by a URL/page name. *(3)*

7. Computer ethics and crime (15 marks)

– (a) Computer ethics = moral principles/guidelines that govern the acceptable use of computers and ICT. *(3)*

– (b) Any 5 of: hacking, software piracy, spreading viruses/malware, cyberbullying, identity theft, phishing, data theft/fraud, unauthorized access — 2 marks each (1 for naming, 1 for brief explanation). *(10)*

– (c) Any 2 of: install antivirus and firewall software, use strong passwords and user accounts, restrict physical access to the lab, keep software updated, train students on safe internet use, regular data backups. *(2)*

SECTION C (35 Marks — best 2 of 3 questions, 17.5 marks each)

8. Impact of ICT in schools (17.5 marks)

– Positive: access to information/research, e-learning materials, faster communication, improved record-keeping, skills for the job market, use of computers for exams/administration. *(≈9)*

– Negative: distraction (social media/games), cost of equipment and maintenance, exposure to harmful/inappropriate content, cyberbullying, reduced physical activity/social interaction, risk of exam malpractice via devices. *(≈9)*

– Mark holistically: award marks for balanced discussion with relevant, well-explained examples rather than a bare list; allow candidate's own valid examples. *(0.5)*

9. Hardware components, faults and fixes (17.5 marks — 3.5 marks per component)

– Motherboard — connects all components together; fault: loose/burnt component causing no boot; fix: reseat components / replace motherboard. *(3.5)*

– Power supply unit (PSU) — converts AC to DC power for components; fault: computer fails to power on; fix: check power cable/replace PSU. *(3.5)*

– Hard disk drive — stores data permanently; fault: 'disk not found' error / slow performance / bad sectors; fix: check cable connection, run disk check, replace disk if damaged. *(3.5)*

– RAM — provides temporary/working memory for running programs; fault: frequent freezing/blue screen; fix: reseat RAM in slot, replace faulty module. *(3.5)*

– Cooling fan — prevents overheating of components; fault: system overheats and shuts down; fix: clean dust, replace fan, apply new thermal paste. *(3.5)*

10. Student registration database design (17.5 marks)

– (a) Any 4 reasonable tables with 2 fields each, e.g.: Students (AdmissionNo, FullName); Classes (ClassID, ClassName); Guardians (GuardianID, GuardianName); Subjects (SubjectCode, SubjectName). 2 marks per correct table+fields = 8 marks; up to 1 mark bonus for a sensible relationship between tables. *(9)*

– (b) Data integrity measures: primary keys to avoid duplicate records, validation rules/data types on fields, referential integrity between related tables (foreign keys), restricting user access/permissions, regular backups. Award marks for any well-explained combination. *(8.5)*

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142/2 INFORMATION AND COMPUTER STUDIES
PAPER 2 — PRACTICAL (Marking Guide)

General note to examiner: Practical work should be assessed on-screen and/or from the candidate's printed output. Award marks for correct process even where the final visual result has minor cosmetic imperfections, provided the required skill was demonstrated.

QUESTION 1 — WORD PROCESSING (30 Marks)

– (a) Letter created, relevant content (~150 words), addressed to parents, mentions Form Four meeting, has date/sender/closing. *(10)*

– (b) (i) Title bold, size 16, centered — 3; (ii) Body justified, size 12 — 3; (iii) Header with school name and footer with page number — 2. *(8)*

– (c) Table with 4 rows × 3 columns, correctly showing a timetable, header row shaded. *(7)*

– (d) File saved with correct name (WORD1_[exam no.]) and successfully printed. *(5)*

QUESTION 2 — SPREADSHEET (35 Marks)

– (a) 6 students' names and marks entered correctly for 3 subjects in a clear table layout. *(8)*

– (b) (i) Total using =SUM() — 5; (ii) Average using =AVERAGE() — 5; (iii) Grade using nested IF() matching the given criteria — 5. *(15)*

– (c) Class average (=AVERAGE of column), highest (=MAX()), lowest (=MIN()) correctly computed for each subject. *(7)*

– (d) Column chart created comparing totals, with a chart title and labelled axes; file saved as EXCEL1_[exam no.]. *(5)*

QUESTION 3 — DATABASE AND WEB DESIGN (35 Marks)

Part A: Database (20 marks)

– (a) Database created and correctly named SCHOOL_[exam no.]. *(2)*

– (b) Table STUDENTS created with all 5 fields, AdmissionNo set as Primary Key, and appropriate data types (e.g. text for names, number for age). *(8)*

– (c) 5 complete and consistent student records entered. *(5)*

– (d) Query correctly filters and displays only Form Four students (criteria: Form = "Four" or "4"). *(3)*

– (e) A usable form created for entering new records, based on the STUDENTS table. *(2)*

Part B: Web design (15 marks)

– (f) HTML page has a <title> and an <h1> heading with the school name. *(4)*

– (g) A descriptive <p> paragraph and an / unordered list of 3 subjects. *(5)*

– (h) A working hyperlink (<a href="https://www.necta.go.tz">) to the NECTA website. *(3)*

– (i) File correctly saved as WEB1_[exam no.].html and displays properly when opened in a browser. *(3)*

Total for Paper 2 = 100 marks. Award part-marks generously for correct method/approach where the final on-screen result is close but not perfect (e.g. correct formula with a minor reference error).